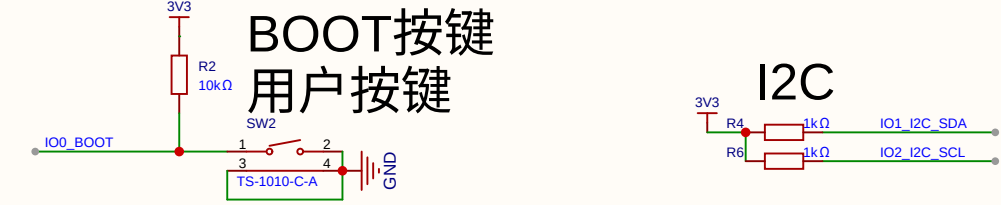
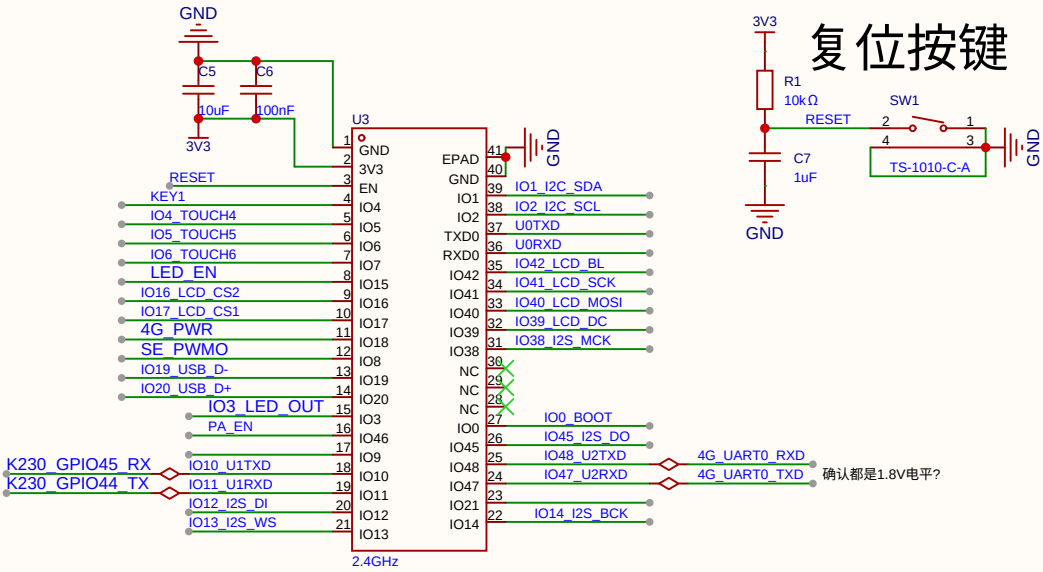
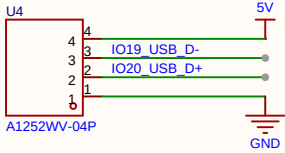


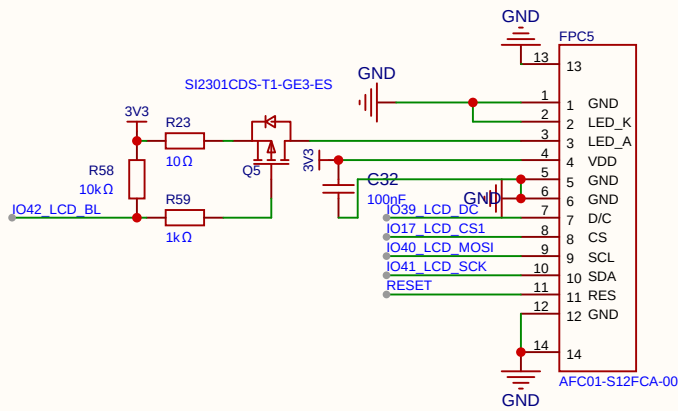
ESP32-S3-WROOM-2-N32R16V



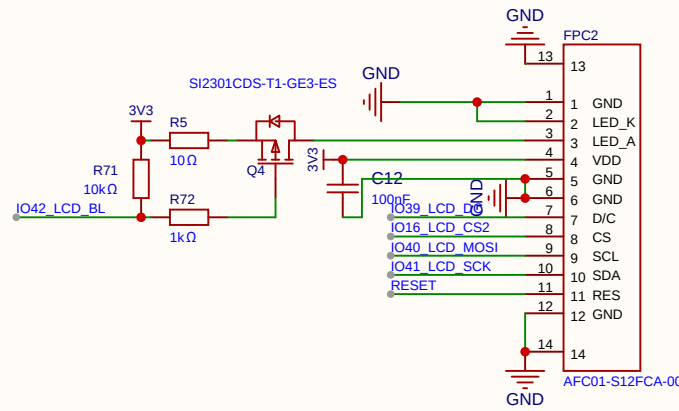
USB摄像头接口



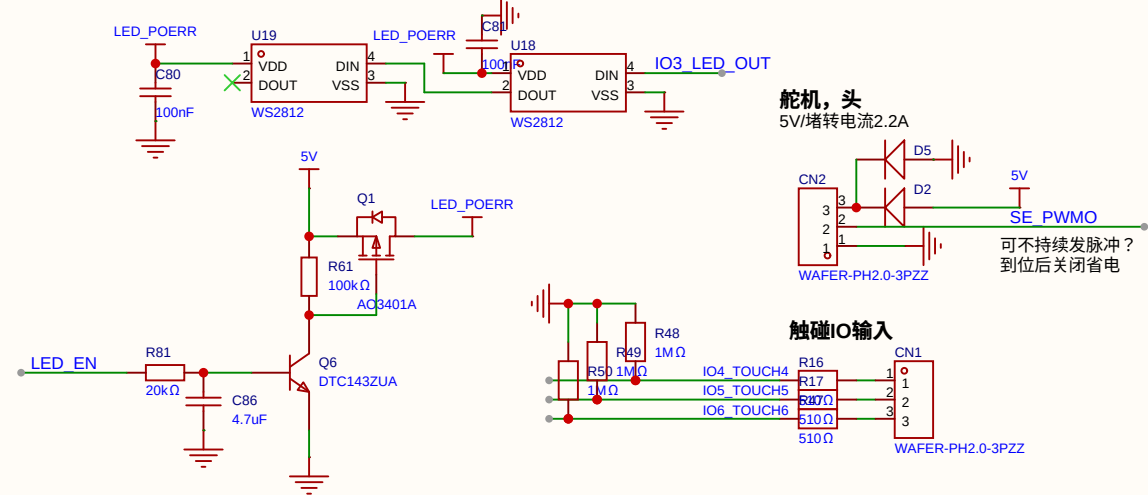
眼睛  
TFT接口



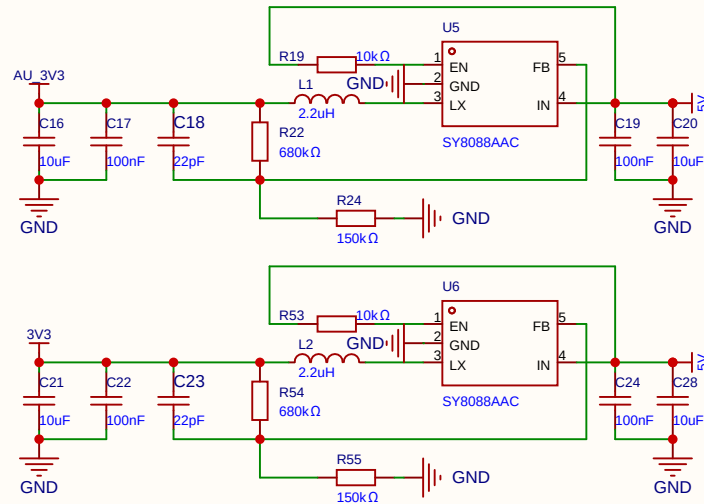
眼睛  
TFT接口



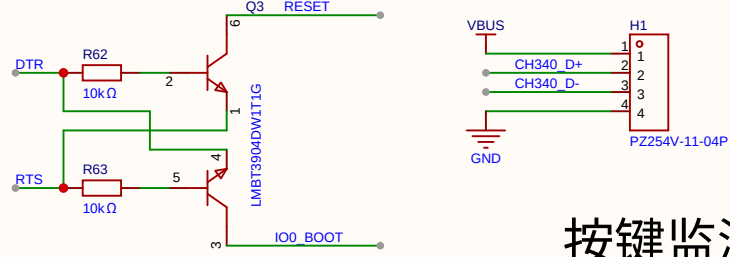
灯带接口



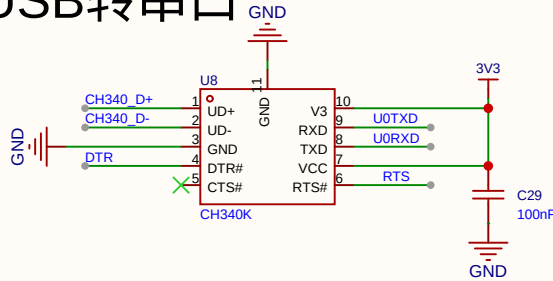
电源



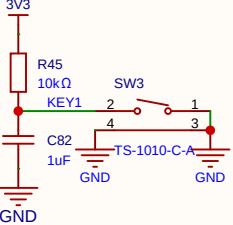
串口自动升级



USB转串口



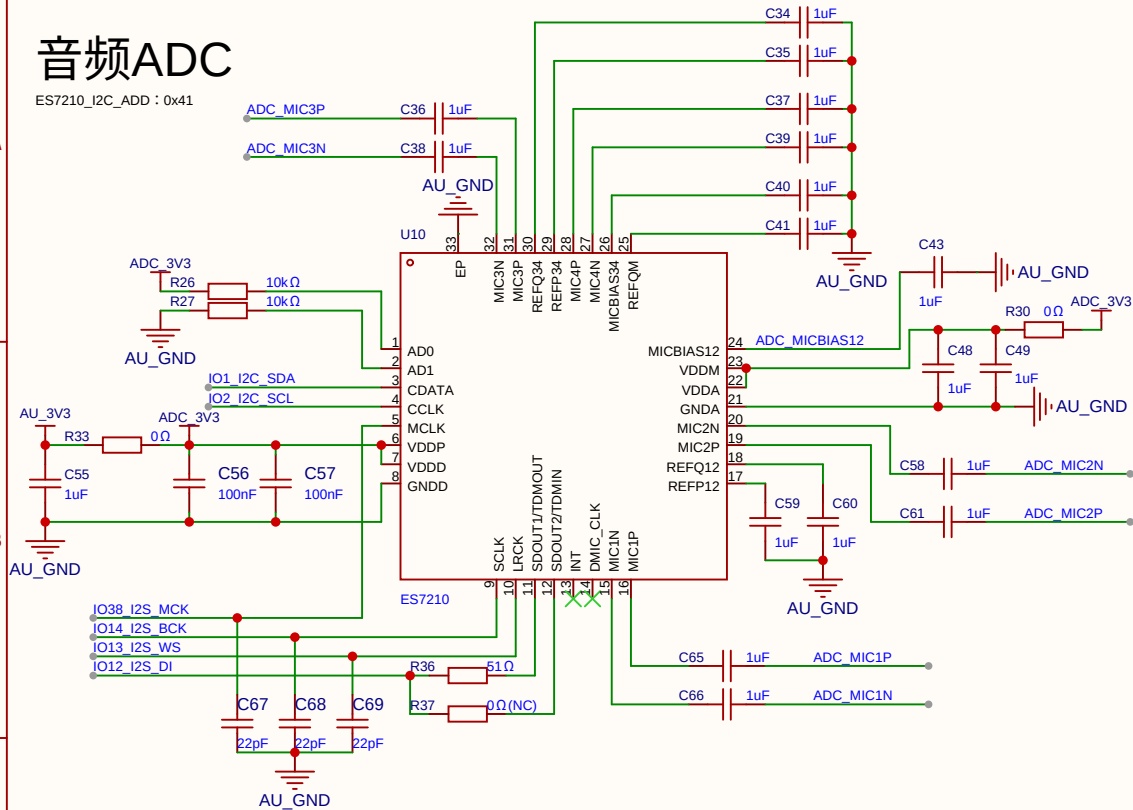
按键监测



|                                                                                       |          |                                 |    |                  |            |
|---------------------------------------------------------------------------------------|----------|---------------------------------|----|------------------|------------|
| 原理图                                                                                   | USB摄像头大板 |                                 |    | 创建日期             | 2026-08-21 |
|                                                                                       |          |                                 |    | 更新日期             | 2026-08-24 |
| 板子                                                                                    | Board1_1 |                                 |    | 图页               | 主控，摄像，眼睛   |
| 绘制                                                                                    |          | New Project_2026-07-19_11-10-43 |    |                  |            |
| 审阅                                                                                    |          |                                 |    |                  |            |
|                                                                                       |          |                                 |    |                  |            |
|                                                                                       |          | 版本                              | 尺寸 | 页    1    共    6 |            |
|  |          | V1.0                            | A4 | 嘉立创EDA           |            |

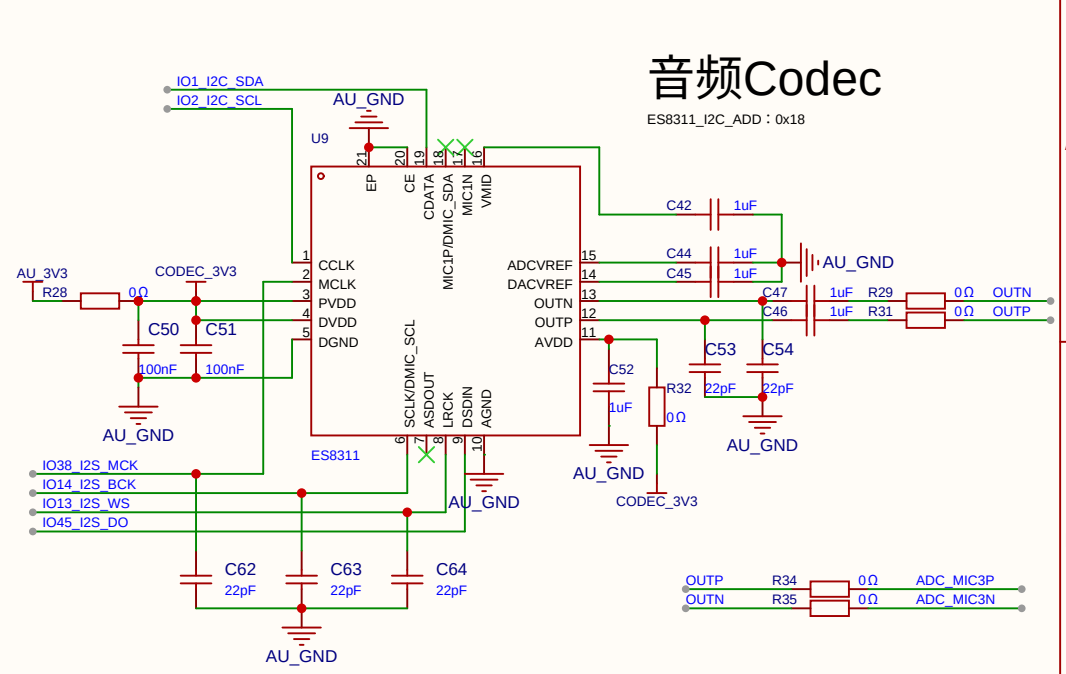
音频ADC

ES7210\_I2C\_ADD : 0x41

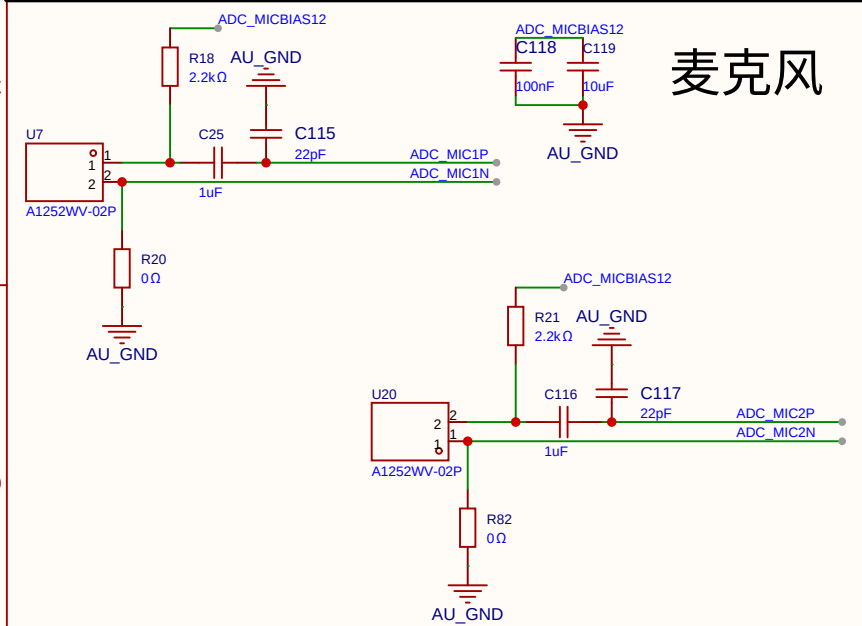


音频Codec

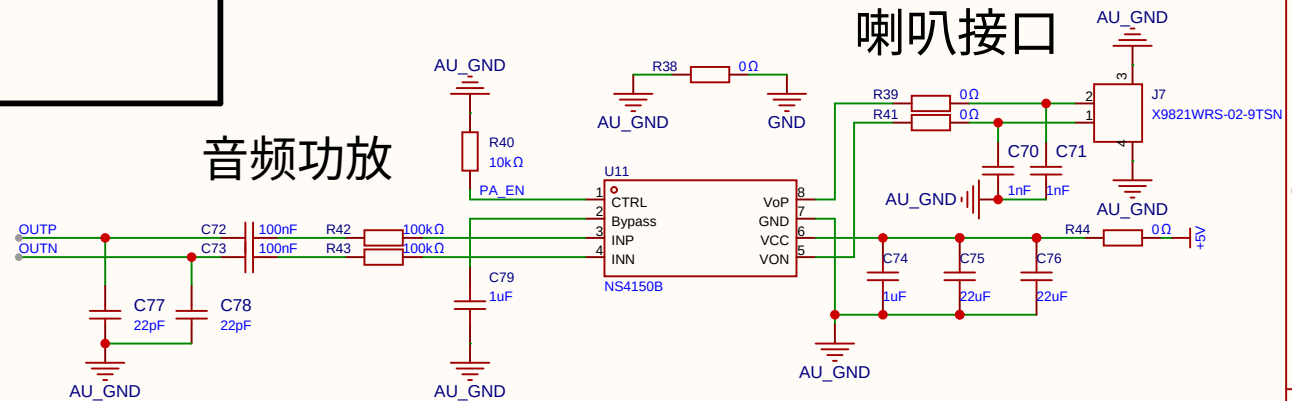
ES8311\_I2C\_ADD : 0x18



麦克风

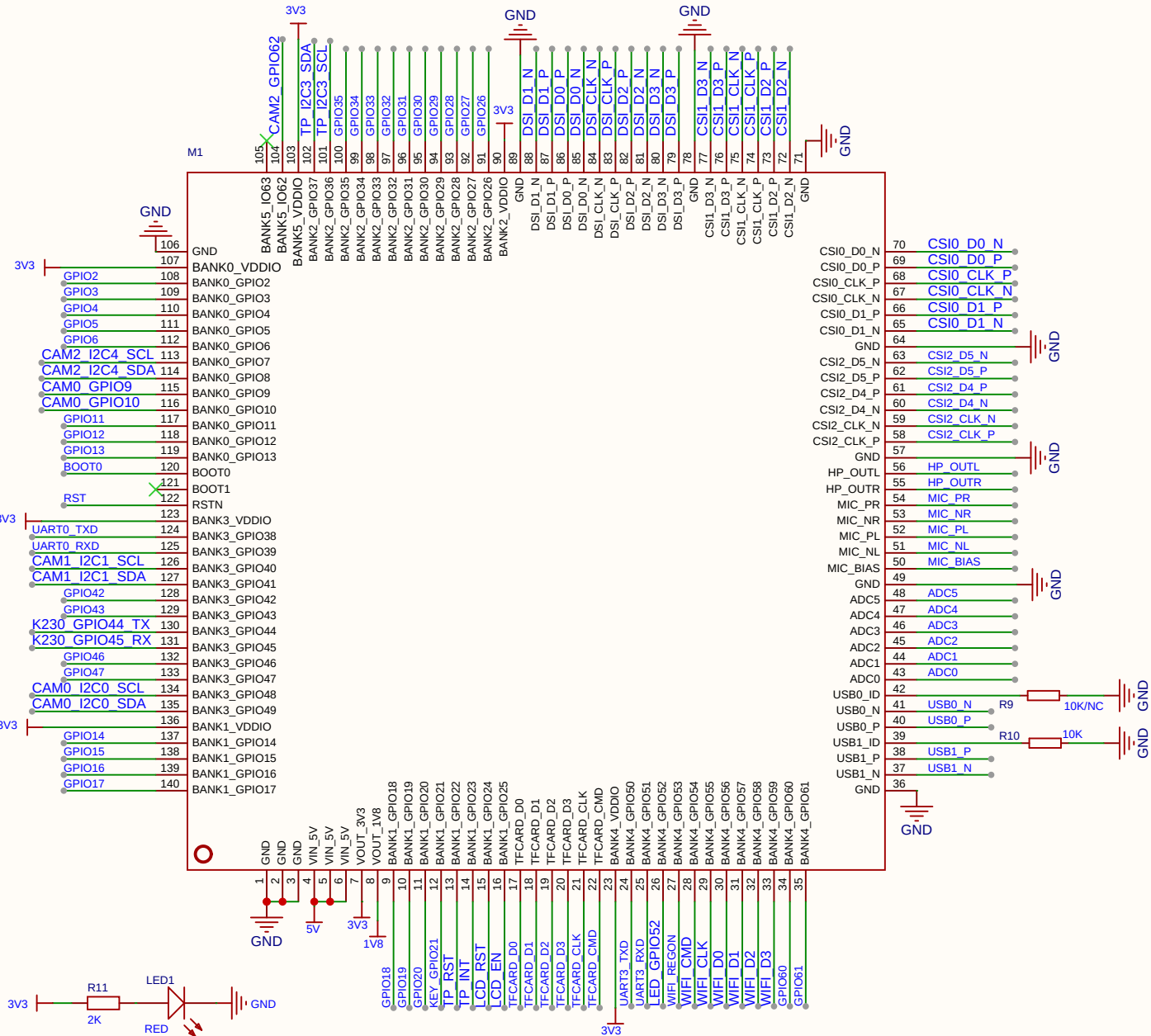


音频功放

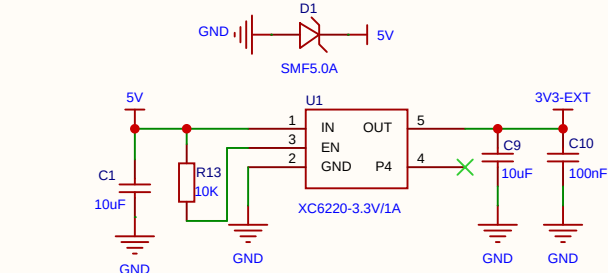


喇叭接口

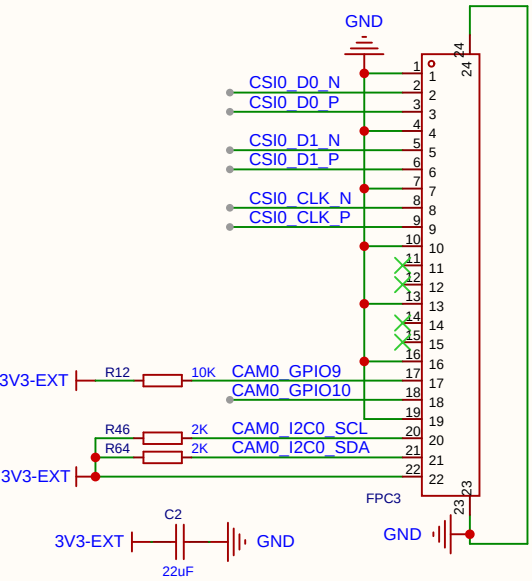
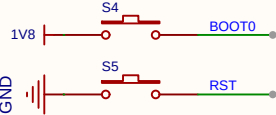
|        |          |      |                                 |            |
|--------|----------|------|---------------------------------|------------|
| 原理图    | USB摄像头大板 |      | 创建日期                            | 2026-08-21 |
| 板子     | Board1_1 |      | 更新日期                            | 2026-08-24 |
| 绘制     |          |      | 图页                              | 咪头, 喇叭     |
| 审阅     |          |      | New Project_2026-07-19_11-10-43 |            |
|        |          | 版本   | 尺寸                              | 页 2 共 6    |
| 嘉立创EDA |          | V1.0 | A4                              | 嘉立创EDA     |



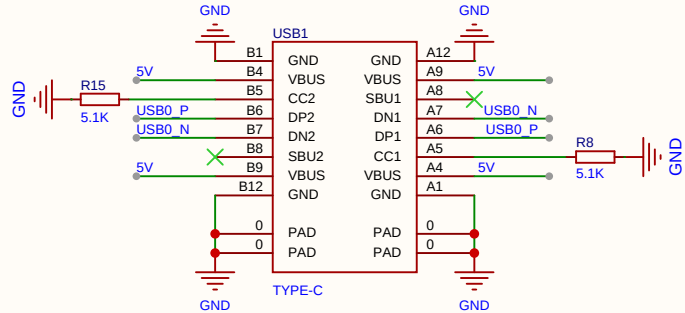
CM-K230



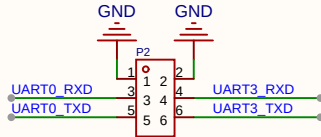
External 3.3V



摄像头



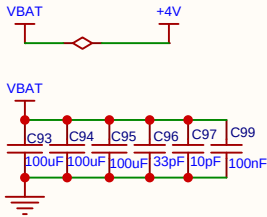
TYPE-C



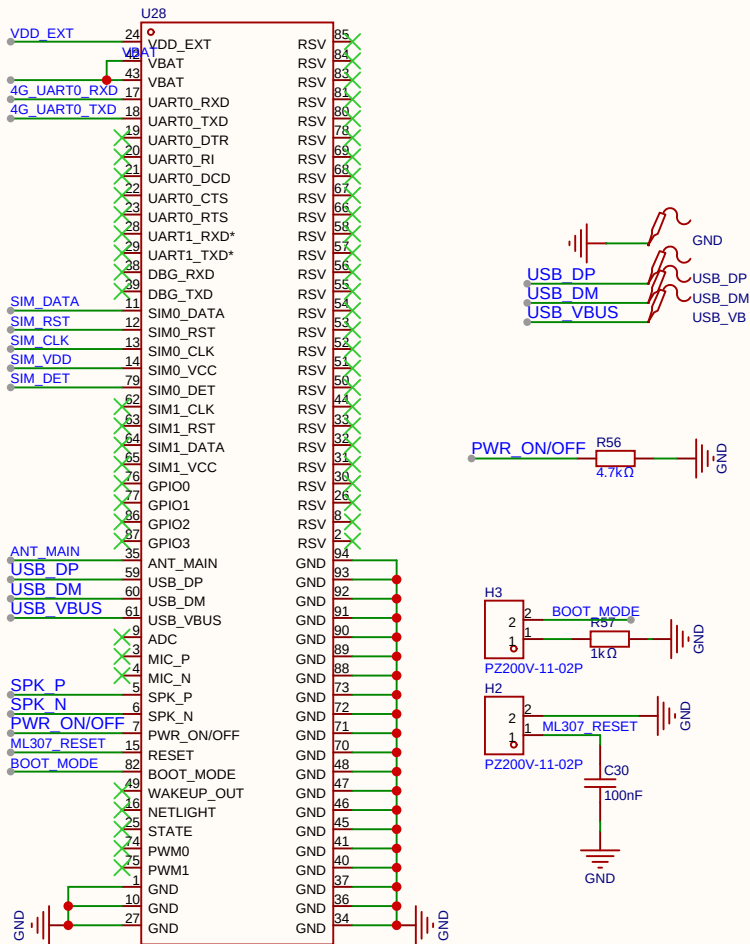
调试

|                                                                                       |          |                                 |    |                  |            |
|---------------------------------------------------------------------------------------|----------|---------------------------------|----|------------------|------------|
| 原理图                                                                                   | USB摄像头大板 |                                 |    | 创建日期             | 2026-08-08 |
|                                                                                       |          |                                 |    | 更新日期             | 2026-08-25 |
| 板子                                                                                    | Board1_1 |                                 |    | 图页               | K230       |
| 绘制                                                                                    |          | New Project_2026-07-19_11-10-43 |    |                  |            |
| 审阅                                                                                    |          |                                 |    |                  |            |
|                                                                                       |          |                                 |    |                  |            |
|                                                                                       |          | 版本                              | 尺寸 | 页    3    共    6 |            |
|  |          | V1.0                            | A4 | 嘉立创EDA           |            |

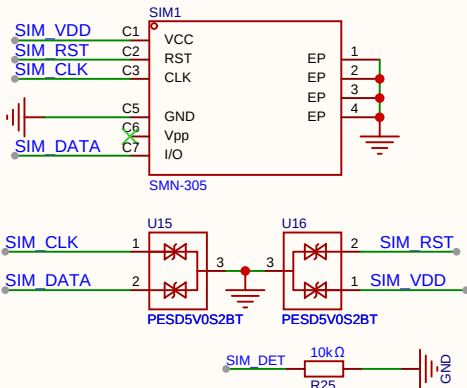
4G电源



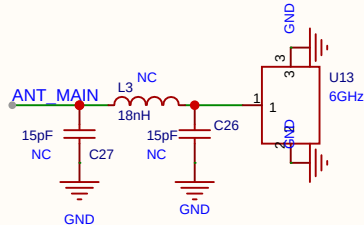
ML307A



SIM

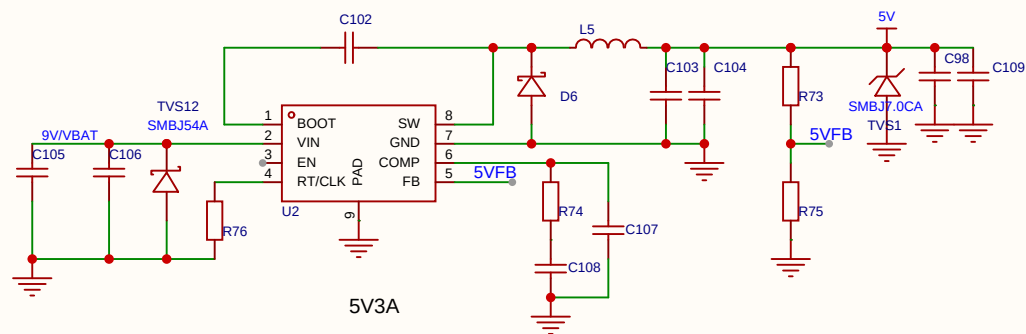


天线

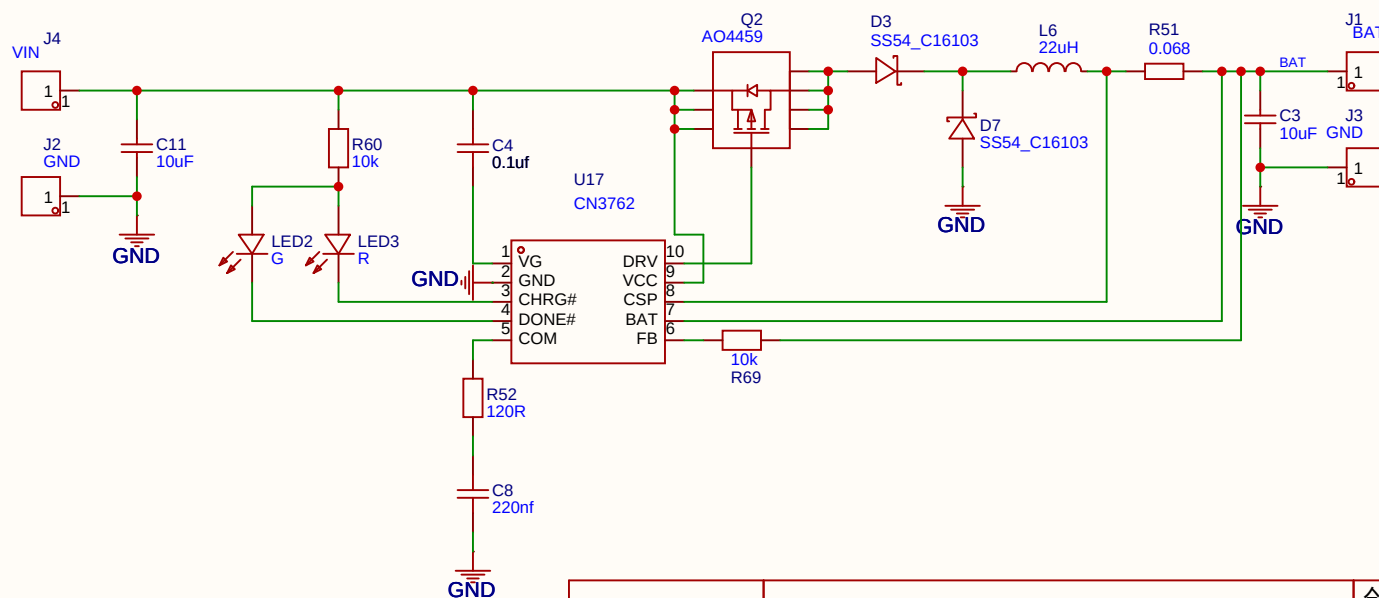
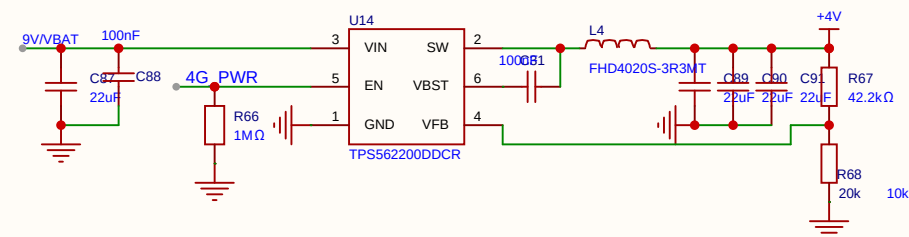


|        |          |      |                                 |            |
|--------|----------|------|---------------------------------|------------|
| 原理图    | USB摄像头大板 |      | 创建日期                            | 2026-07-30 |
| 板子     | Board1_1 |      | 更新日期                            | 2026-08-24 |
| 绘制     |          |      | 图页                              | 4G模块       |
| 审阅     |          |      | New Project_2026-07-19_11-10-43 |            |
|        |          | 版本   | 尺寸                              | 页 4 共 6    |
| 嘉立创EDA |          | V1.0 | A4                              | 嘉立创EDA     |

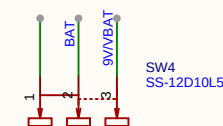
## 整版5V供电



## 4G模块供电



## 总电源开关



|                                                                                      |          |                                 |    |        |            |  |   |   |   |
|--------------------------------------------------------------------------------------|----------|---------------------------------|----|--------|------------|--|---|---|---|
| 原理图                                                                                  | USB摄像头大板 |                                 |    | 创建日期   | 2026-08-21 |  |   |   |   |
|                                                                                      |          |                                 |    | 更新日期   | 2026-08-24 |  |   |   |   |
| 板子                                                                                   | Board1_1 |                                 |    | 图页     | 电池         |  |   |   |   |
| 绘制                                                                                   |          | New Project_2026-07-19_11-10-43 |    |        |            |  |   |   |   |
| 审阅                                                                                   |          |                                 |    |        |            |  |   |   |   |
|                                                                                      |          |                                 |    |        |            |  |   |   |   |
|                                                                                      |          |                                 |    |        |            |  |   |   |   |
|                                                                                      |          | 版本                              | 尺寸 | 页      |            |  | 5 | 共 | 6 |
|  |          | V1.0                            | A4 | 嘉立创EDA |            |  |   |   |   |

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| A |   |   |   |   | A |
| B |   |   |   |   | B |
| C |   |   |   |   | C |
| D |   |   |   |   | D |
| 1 | 2 | 3 | 4 | 5 | 6 |

|                                                                                      |          |                                 |      |            |
|--------------------------------------------------------------------------------------|----------|---------------------------------|------|------------|
| 原理图                                                                                  | USB摄像头大板 |                                 | 创建日期 | 2026-07-30 |
|                                                                                      |          |                                 | 更新日期 | 2026-08-21 |
| 板子                                                                                   | Board1_1 |                                 | 图页   | P2         |
| 绘制                                                                                   |          | New Project_2026-07-19_11-10-43 |      |            |
| 审阅                                                                                   |          |                                 |      |            |
|                                                                                      |          |                                 |      |            |
|                                                                                      |          | 版本                              | 尺寸   | 页 6 共 6    |
|  |          | V1.0                            | A4   | 嘉立创EDA     |